

LESSON 8-4 • SESSION 2 • PRACTICE • FORM A

Representing Inequalities

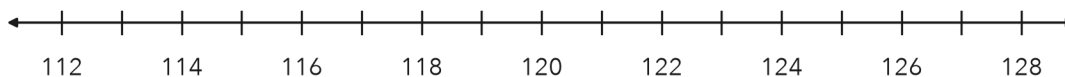
Learning target – I can represent an inequality with a set of solutions and describe them. • 6.EE.B.8

REMEMBER

- **Closed** (filled) circle: the boundary counts ($\leq$, $\geq$).
Open circle: it does not ($<$, $>$).
- Arrow **left** for less than. Arrow **right** for greater than.
- Check the story: people and tickets come only in **whole numbers**.

WORKED EXAMPLEAn elevator allows no more than 8 people. Show the solutions for p .**Step 1** Write the rule: $p \leq 8$.**Step 2** "No more than" includes 8, so draw a closed circle at 8 and an arrow left.**Step 3** People come in whole numbers: 0, 1, 2, ..., 8.**Answer** $p \leq 8$, whole numbers 0 to 8**WORDS** number line (*recta numérica*) • open circle (*círculo abierto*) • closed circle (*círculo cerrado*)**PRACTICE**

show your work • label units

1. Write the inequality for each graph. Use x .**a.** open circle at 4, arrow right _____**b.** closed circle at -1 , arrow left _____**c.** closed circle at 6, arrow right _____**d.** open circle at 0, arrow left _____**2. Meeting Room.** A meeting room may hold at most 120 people at one time.**a.** Write an inequality for n , the number of people. _____**b.** Graph it on the number line.**3. Fish Tank.** The graph shows the safe water temperatures, t , for a fish tank. Write the inequality. Is 78 degrees a solution?**Hint** Look at the circle at 78. Is it open or closed?**Answer** _____**4. Locker Bag.** A bag must be shorter than 22 inches to fit in a locker. Write an inequality for h . Name two solutions and one non-solution.

Show your work.

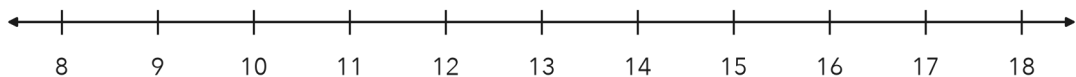
Answer _____**5. Lab Station.** A lab station may hold at most 5 students. Write an inequality for s . Is 5 a solution?**Hint** "At most" includes the boundary.

Show your work.

Answer _____

6. **Parade Float.** A float must be shorter than 14 feet to pass under a bridge.

Hint "Shorter than" leaves the boundary out.



- a. Write an inequality for h . _____
- b. Graph it on the number line.
7. **Name Solutions.** For $n > 7$, name three solutions and one number that is not a solution.

Show your work.

Answer _____

8. **Delivery Limit.** A van's load, w , must be at least 50 pounds and at most 900 pounds.

- a. Write both inequalities. _____
- b. Name three loads that fit both rules. _____
- c. Why is 900 pounds allowed but 901 pounds is not? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Zuri's Work.** A sign says: "Carry no more than 6 items into the fitting room." Zuri wrote $i < 6$. Explain her mistake. Write the correct inequality.

TEST PRACTICE

choose the best answer

10. Which value is a solution of $n > 5$?

- A** 3 **B** 5 **C** 4.9 **D** 6

11. Which value is NOT a solution of $x \geq 4$?

- A** 4 **B** 3.5 **C** 7 **D** 10

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

h < 14 L	s < 5 X	w > 50 J	t < 78 I	w ≥ 50 E	n ≤ 120 C	t ≤ 78 Z	s ≤ 5 C	h < 22 R	n < 120 Q
		#2a	#3	#4	#5	#6a	#8a		

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____